

IMO Training – Algebraic Number Theory

Field Extensions

Longtin Chan
longtinclt@gmail.com

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Introduction

Notation. Let $*$: $\{a\} \times S \rightarrow R$ be a binary operation. We define $a * S = \{a * s : s \in S\}$ and $S * a = \{s * a : s \in S\}$.

Definition (Group). A *group* is a triple $(G, \cdot, e)$, where G is a set, $\cdot : G \times G \rightarrow G$ is a function and $e \in G$ is an element such that:

1. For all $a, b, c \in G$, we have $(a \cdot b) \cdot c = a \cdot (b \cdot c)$.
2. For all $a \in G$, we have $a \cdot e = e \cdot a = a$.
3. For all $a \in G$, there exists $a^{-1} \in G$ such that $a \cdot a^{-1} = a^{-1} \cdot a = e$.

Definition (Subgroup). If $(G, \cdot, e)$ is a group and $H \subset G$ is a subset, then H is a *subgroup* if

1. $e \in H$,
2. $a, b \in H$ implies $a \cdot b \in H$,
3. $\cdot : H \times H \rightarrow H$ makes $(H, \cdot, e)$ a group.

Definition (Abelian group). A group G is *abelian* if $a \cdot b = b \cdot a$ for all $a, b \in G$.

Definition (Group Homomorphism). For two groups $(G, *)$ and $(H, \times)$, A function $\phi : G \rightarrow H$ is a *group homomorphism* if $\phi(g * h) = \phi(g) \times \phi(h)$ for all $g, h \in G$.

Definition (Ring). A *ring* is a quintuple $(R, +, \cdot, 0, 1)$, where R is a set, $+, \cdot : R \times R \rightarrow R$ are binary operations and $0, 1 \in R$ are elements such that:

1. $(R, +, 0)$ is an abelian group.
2. The operation $\cdot : R \times R \rightarrow R$ satisfies associativity $(a \cdot b) \cdot c = a \cdot (b \cdot c)$ and identity $1 \cdot r = r \cdot 1 = r$.
3. Multiplication distributes over addition, i.e.

$$\begin{aligned}r_1 \cdot (r_2 + r_3) &= (r_1 \cdot r_2) + (r_1 \cdot r_3) \\(r_1 + r_2) \cdot r_3 &= (r_1 \cdot r_3) + (r_2 \cdot r_3)\end{aligned}$$

Remark. The set of rational integers $\mathbb{Z}$ is a ring.

Definition (Commutative ring). A ring R is *commutative* if $a \cdot b = b \cdot a$ for all $a, b \in R$.

Definition (Subring). Let $(R, +, \cdot, 0, 1)$ be a ring, and $S \subseteq R$ be a subset. We say S is a *subring* of R if $0, 1 \in S$, and the operations $+, \cdot$ make S into a ring in its own right. In this case we write $S \leq R$.

Definition (Ring Homomorphism). Let R, S be rings. A function $\phi : R \rightarrow S$ is a *ring homomorphism* if it preserves everything we can think of, i.e.

1. $\phi(r_1 + r_2) = \phi(r_1) + \phi(r_2)$,
2. $\phi(0_R) = 0_S$,
3. $\phi(r_1 \cdot r_2) = \phi(r_1) \cdot \phi(r_2)$,
4. $\phi(1_R) = 1_S$.

Definition (Ring Isomorphism, Monomorphism and Automorphism). If a homomorphism is an injection, we call it an *monomorphism*. If a homomorphism $\phi : R \rightarrow S$ is a bijection, we call it an *isomorphism*. An isomorphism of R with itself is called an *automorphism* of R .

Definition (Unit). An element $u \in R$ is a *unit* if there is another element $v \in R$ such that $u \cdot v = 1$.

Definition (Field). A *field* is a non-zero ring F where every non-zero $u \in F$ is a unit.

Definition (Associates). $a, b \in R$ are *associates* if $a = bc$ for some unit c .

Definition (Irreducible). $a \in R$ is *irreducible* if $a \neq 0$, a is not a unit, and if $a = xy$, then x or y is a unit.

Definition (Polynomial). Let R be a ring. Then a *polynomial* with coefficients in R is an expression

$$f = a_0 + a_1X + a_2X^2 + \cdots + a_nX^n$$

with $a_i \in R$.

Definition (Polynomial ring). We write $R[X]$ for the ring formed by the set of all polynomials with coefficients in R . The operations are performed the usual way.

Definition (Vector space). An F -*vector space* is an (additive) abelian group V , a field F , together with a function $F \times V \rightarrow V$, written $(\lambda, \mathbf{v}) \mapsto \lambda\mathbf{v}$, such that

1. $\lambda(\mu\mathbf{v}) = \lambda\mu\mathbf{v}$ for all $\lambda, \mu \in F$, $\mathbf{v} \in V$ (associativity)
2. $\lambda(\mathbf{u} + \mathbf{v}) = \lambda\mathbf{u} + \lambda\mathbf{v}$ for all $\lambda \in F$, $\mathbf{u}, \mathbf{v} \in V$ (distributivity in V)
3. $(\lambda + \mu)\mathbf{v} = \lambda\mathbf{v} + \mu\mathbf{v}$ for all $\lambda, \mu \in F$, $\mathbf{v} \in V$ (distributivity in F)
4. $1\mathbf{v} = \mathbf{v}$ for all $\mathbf{v} \in V$ (identity)

We always write $\mathbf{0}$ for the additive identity in V , and call this the identity. By abuse of notation, we also write 0 for the trivial vector space $\{\mathbf{0}\}$.

Definition (Generating set). Let R be a ring and let $V \subset U \subset R[X]$. V is a *generating set* of U if for every $f \in U$, there exist $f_1, \dots, f_n \in V$ and $r_1, \dots, r_n \in R$ such that $f = r_1f_1 + \cdots + r_nf_n$.

Definition (Basis). Let R be a ring and let $V \subset U \subset R[X]$. V is a *basis* of U if V is a generating set of U and if V is linearly independent, i.e. for every $f_1, \dots, f_n \in V$, $r_1f_1 + \cdots + r_nf_n = 0$ implies $r_1 = \cdots = r_n = 0$.

Example. Prove that the following numbers are irrational:

1. $\sqrt[3]{1001}$
2. $\sqrt{2} + \sqrt{3} + \sqrt{6}$
3. $\sqrt{2} + \sqrt[3]{3}$

Lemma (Gauss Lemma). Let f be a polynomial over $\mathbb{Z}$ that is irreducible over $\mathbb{Z}$. Then f is also irreducible over $\mathbb{Q}$.

Theorem (Eisenstein's Criterion). Let $f(t) = \sum_{k=0}^n a_k t^k$ be a polynomial over $\mathbb{Z}$. Suppose there is a prime p such that

1. $p \nmid a_n$
2. $p \mid a_i$ for all $0 \leq i \leq n-1$
3. $p^2 \nmid a_0$

Then f is irreducible over $\mathbb{Q}$.

Example. Determine whether the following polynomials are irreducible in $\mathbb{Q}[X]$:

1. $X^4 - 2X + 2$
2. $X^4 + 18X^2 + 24$
3. $X^4 + 4$
4. $X^{p-1} + X^{p-2} + X^{p-3} + \cdots + X + 1$, where p is a prime

Definition (Field Extension). A *field extension* is an inclusion of a field $K \subseteq L$, where K inherits the algebraic operations from L . We also write this as L/K . Alternatively, we can define this by a injective homomorphism $K \rightarrow L$. We say L is an *extension* of K , and K is a *subfield* of L .

Definition. If $K \subseteq L$ is a field extension and $Y \subseteq L$ is a subset, then the subfield of $\mathbb{C}$ generated by $K \cup Y$ is written $K(Y)$ and is said to be obtained from K by *adjoining* Y . This field is defined by the smallest subfield containing K and Y .

Definition (Simple Extension). A field extension $K \subseteq L$ is *simple* if $L = K(\alpha)$ for some $\alpha \in L$.

Example. Prove that $\mathbb{Q}(\sqrt{2}, \sqrt{3}) = \mathbb{Q}(\sqrt{2} + \sqrt{3})$

Example. Rationalise:

1.
$$\frac{1}{1 + \sqrt{2} - \sqrt{3} + \sqrt{6}}$$

2.
$$\frac{1}{\sqrt[3]{4} + \sqrt[3]{2} + 3}$$

Definition (Algebraic and Transcendental). Let K be a subfield of $\mathbb{C}$ and let $\alpha \in \mathbb{C}$. Then α is *algebraic* over K if there exists a nonzero polynomial $P(X) \in K[X]$ such that $P(\alpha) = 0$. Otherwise, α is *transcendental* over K .

Definition (Minimal Polynomial). Let $K \subseteq L$ be a field extension, and suppose that $\alpha \in L$ is algebraic over K . Then the *minimal polynomial* of α over K is the unique monic polynomial $\mu_\alpha(X) \in K[X]$ of minimal degree such that $\mu_\alpha(\alpha) = 0$.

Definition (Algebraic Extension). A field extension $K \subseteq L$ is *algebraic* if every element of L is algebraic over K .

Example. Find the minimal polynomial with rational coefficients of the following numbers:

1. $\sqrt{3} + \sqrt{5}$
2. $\sqrt[3]{2} + \sqrt[3]{4}$
3. $\cos \frac{2\pi}{7}$

Proposition. If α is algebraic over $K \subseteq \mathbb{C}$, then μ_α is irreducible over K . It divides every polynomial P with $P(\alpha) = 0$.

Proposition. For any subfield $K \subseteq \mathbb{C}$ and P any irreducible monic polynomial over K , $P = \mu_\alpha$ for some $\alpha \in \mathbb{C}$ that is algebraic over K .

Proposition. A field extension $K \subseteq L$ is finite if and only if L is algebraic over K and there exist finitely many elements $\alpha_1, \alpha_2, \dots, \alpha_s \in L$ such that $L = K(\alpha_1, \alpha_2, \dots, \alpha_s)$.

Definition (Degree of Extension). The *degree* of L over K is $[L : K]$ is the dimension of L as a vector space over K . The extension is *finite* if the degree is finite.

Theorem (Tower Law). If $K \subseteq L \subseteq M$ are subfields of $\mathbb{C}$, then $[M : K] = [M : L][L : K]$.

Proposition. Let $K(\alpha) : K$ be a simple extension. If it is transcendental, then $[K(\alpha) : K] = \infty$. If it is algebraic, then $[K(\alpha) : K] = \deg \mu_\alpha$.

Example. Compute the following:

1. $[\mathbb{Q}(\sqrt{2}, \sqrt{3}) : \mathbb{Q}]$
2. $[\mathbb{Q}(\sqrt{2} + \sqrt[3]{2}) : \mathbb{Q}]$